

1 Explain types of medium?

1. Radio frequency 2. Electrical frequency 3. Light Signal (FOC)

2 Types of network communication?

Two types 1. Wired and 2. Wireless

3. Write down PC and Switch 8 pins in order?

4. Color code of T568B? RJ45 PC to Switch use Straight cable

T568B to T568B - Same pin orders both ends

1. White Orange - Transmit Data +

2. Orange - Transmit Data -

3. White Green - Receive Data +

4. Blue - ~~Receive Data -~~ POE Volt + Positive

5. White Blue - POE Volt - Negative

6. Green - Receive Data -

7. White Brown - Dummy - negative

8. Brown - Dummy + Positive

5 PC to PC Cable Crimping type?

PC to PC - we can use Cross cable - PC1 - T568A - PC2 - T568B

6 Router to Switch cable crimping type?

Router to Switch we can use Straight cable Both ends Same

7 Explain Auto-MDIX function? Automatic Medium Dependent Interface

Crossover - This feature that automatically detects whether a connected cable is a Straight (or) Cross cable.

8 Explain function of wired and wireless nic?

Network Interface Card - It connects your device to the network

1. Wired NIC (Ethernet card): Using RJ45 port,

Converts computer data (digital) into electrical signals and sends through LAN cable and receives electrical signals and converts to computer data.

2. Wireless NIC (WiFi card / WLAN card): Using Antenna

Converts computer data into radio waves / RF signals and sends into Air. And receives radio waves and converts back to data.

9 What is POE? Power over Ethernet. POE is basically sending electricity + data through the same Ethernet cable.

10 ~~What is~~

10. POE Switch VS POE Injector?

Both give Power + Data LAN cable, POE Injector - It just an adapter that adds power to your ethernet cable.

11. Write IP Parameter of 181.0.20.0?

Network ID - - 181.0.0.0

First IP - 181.0.0.1

Last IP - 181.0.255.254

Broadcast IP - 181.0.255.255

Subnetmask - 255.255.0.0

Prefix - 16

Total no of IP - 65534 + 2 = 65536

Previous IP - 181.0.20.255

Next IP - 181.0.20.1

12. Why Subnet mask?

Subnet mask is tells which part is N/w and which part is host.

13. What is RDP? Remote Desktop Protocol

It lets you control another computer from your computer. Both PC's need internet (or) same N/w and RDP must be Enabled.

14. Explain netbios?

Find PC's ~~find~~ in LAN by Name not by IP.

15. Explain OSI Layer when we ping 70.10 to 70.20?

7. Application - Echo data generated.

6. Presentation - Echo + Format

5. Session - Echo + Format + Time

4. Transportation - Echo + Format + Time + TCP/UDP

3. Network - S. IP - 70.10, D. IP - 70.20

If same network no need gateway

2. Data link - Need Target MAC

ARP Request - S. IP, S. mac, T. IP, T. MAC

Sender IP : 70.0.0.10

Sender Mac : 00A

Target IP : 70.0.0.20

Target Mac : Broadcasting

1. Physical connection: Electrical signal sent to switch

Switch (Layer 2): Check MAC Table, If not MAC then forward to port where 20 is connected

1. Physical: Receives Signals

2. D-Link: Check MAC is mine

3. N/w: Check this IP is mine

2. D-Link: ARP Reply

Sender IP: 70.0.0.20

Sender mac: 00B

Target IP: 70.0.0.10

Target mac: 00A

1. Physical: Send back

If it was different N/w

It would send it to gateway instead of direct ARP

16. When we need Switch device?

We need Switch when you have more than 2 devices in same N/w
It works on Layer 2 - Using MAC Address.

17. When we need Router device?

Switch connects same N/w, Router connects different N/w

18. Switch packet flow?

If data comes to Switch, Switch 1st do the Address learning, Note the Sender IP & Sender Mac, then Target IP & Mac is not in MAC table it broadcasting the Request.

19. Router Packet Flow?

It is use to connect different N/w, It has Routing table, It have 3 Info - 1. N/w ID, 2. Prefix, 3. Interface Name Directly connected N/w.

20. What is directly connected Networks?

N/w that is Physically connected to Router's own port.

21. What is the command to enable telnet?

Router(config)# Username: Telnet Password: Telnet

Router(config)# Line vty 0 4 Router(config)# Transport input telnet

Router(config)# Login Local Router(config)# Exit

22. How to set console port authentication?

Router(config)# Line console 0, Router(config)# Password console

Router(config)# Login Local Router(config)# Exit

23. Write System route table of 129.0.129.0?

129.0.129.0 0.0.0.0 129.0.129.10

129.0.129.10 0.0.0.0 129.0.0.1

24. Explain destination host unreachable?

When PC don't have route for the destination, The result will be destination host unreachable.

25. Explain request timed out?

When PC have ^{route for} destination IP and it will send ARP request out but, it don't have Reply, So if there some issue, it will show Time request out.

26. Explain Importance of default gateway?

A default gateway is Layer 3, It helps send data to different Network.

27 What is Static Route?

A static route is a route we manually entered our devices. It doesn't learn it automatically.

28 Write a command & Syntax of Static Route?

IP Route -> N/W 5.0.0.0 Subnetmask 255.0.0.0 Interface 5.0.0.10

29 Explain Loadsharing Type in Cisco router?

Two types 1. Per destination 2. Per Packet

1. Per destination - Router sends all packets for same source

2. Per Packets - IP loadsharing per packet, Router sends packet one on path one, packet 2 on path 2

30 Explain when loadsharing will happen in Cisco Router?

Two (or) more paths to send traffic to the same destination at the same time, instead of using just one path.

31 What is the command to change loadsharing?

If the IP have multiple routes for destination, we need to see the highest matching bits. If it's also same it moves to 15.

32 Explain Layer 3 loop in cloop?

Layer 3 loop - Routing loop occurs when an IP packets get trapped in a circular path between routers (or) layer 3 switches, passing back and forth endlessly until a safety mechanism stops it.

33 Explain TTL and its importance? (Time To Live)

It acts as an expiration timer or a hop limit for data moving across a network. Each time the packets pass through a router (a hop) the router reduces the TTL number by 1. If the TTL drops to 0 before reaching its destination, the router deletes the packet and sends an error message.

Importance: It prevents routing loop.

34 Explain arp in layer 2 and layer 3?

L3 is IP is logical address for end to end, but switch and actual physical delivery needs Mac address (L2). Router can't forward without Mac. L3 part -> Ping IP -> PC check is it in subnet, if yes, it needs mac, if no - it needs gateway. L2 part ARP Req -> Broadcasting 1st time -> Receive ARP Reply -> L2 is Unicast back to PC -> then MAC TABLE noted address

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~~Networks~~

35 What is MTU? [Maximum Transmission Unit]

The maximum size of a packet that an interface can send in one go, in bytes. Ethernet bytes - 1500 bytes

If Bit is 0 it means Off, If Bit is 1 it means ON

When we send the data one PC to another PC, it will go to 1500 bytes only, If the data is more than 1500 bytes, It will go to Segment wise.

36 What is Segment, SYN-NO, window size?

Layer 4 concepts, 1. Segment - Data broken into small pieces by TCP, Each piece = Segment - Per Size 1460 bytes.

2. SEQNO - It keeps order. Every byte gets a number

PC → SYN → Say Hi and mention its number, ~~2~~

Server → SYN-ACK → Reply Hi and mention its number

PC → ACK → Got it let's start the data

3. Window Size - How much data Server/Receiver can accept at once without ACK. It's flow control

37 Explain importance of 3 way Handshake?

It's compulsory because TCP needs to agree on 3 things before sending Real data.

38 Difference between Linux and windows?

Linux is open source, free access, It's just a kernel people build different versions on top of it Ubuntu, Fedora, mint.

Windows is proprietary, made by Microsoft you pay for it, you get it as-is

39 What is kernel?

kernel is the heart of the all OS, your apps can't talk directly to hardware. They talk to the kernel and the kernel does the actual work.

Linux uses the Linux kernel - open source - made by Linus Torvalds.

40 Difference between client and server OS?

Client OS: is - OS made for user is use to daily basis

Server OS: - OS made to serve thousands of clients.

Client OS is for using, Server OS is for serving.

41 Explain about webservice?

A webservice is just a way for two computers to talk to each other over the internet. ~~You know client~~
~~and~~ If we send `www.hotstar.com` in one browser, ~~to server~~
 Server sends you a full webpage to look at.

42 Explain importance of DNS?

DNS - Domain Name System, It's the notebook of the internet. Computers don't understand names like `www.google.com`, they only understand numbers (or) IP, we can't remember the numbers, so this DNS helps the translator job.

43 What is FQDN? Fully Qualified Domain Name

It is the complete, full address of a computer on the internet. No shortcuts. Example: `www.google.com`

• `.com` - Top level Domain (TLD)

• `google` - Domain name registered

`www` - Host Name

44 Explain packet flow of `labs.surennetwork.com`?

When we enter `labs.surennetwork.com` in browser on PC 1, DNS server receives packet on port 53, it checks its zone database. Since it has in zone `surennetwork.com`. Then DNS server response packet flow is it send source port: 53 and destination to PC 1 IP address. Reply is `labs.surennetwork.com` in a window OS server IP address with TTL value and authoritative answer flag = 1, then PC 1 receives and stores this in its DNS cache.

45 What is port number?

A port number is a 16 bit logical number used to identify a specific process (or) service running on a computer. An IP address gets the packet to the correct computer, but the port numbers get the packet to the correct application inside that computer.

Ex-Port number: Telnet-23, SSH-22, DNS-53

46 What is port number of NTP IMAP POP3?

Network Time Protocol: Port 123 UDP - Used to time Synchronization between Systems.

Internet Message Access Protocol: Port 143 TCP, Normal Port 993 TCP Secure IMAP - Used to read Emails while keeping them on the mail server. You can access mail from multiple devices.

Post office Protocol v3: Port: 110 TCP - Normal Port: 995 TCP - Secure POP3 - Used to download emails from server to local PC and usually deletes from server. Single device access.

47 Explain DHCP and DHCP process?

Dynamic Host Configuration Protocol: It's application layer protocol that automatically assigns IP add and other N/w configuration. It works on DORA Process - Discover Offer Request Acknowledge.

It works Port 67 (server) Port 68 (client)

Discover: PC need an IP so it is broadcast DHCP

offer: DHCP server checks its IP pool those IP's

Request: PC may receives offer from multiple DHCP servers, it select one and says it wants IP

Acknowledge: DHCP server finalizes and gives the IP officially - (unicast)

It's called DORA Process, If no DHCP server windows assigns APIPA IP: 169.254.X.X

48 How DHCP works on same and different network?

Same N/w - DORA process works directly because Discover is broadcast.

PC sends DHCP Discover to 255.255.255.255:-

this is floods to all ports in same VLAN

DHCP server in same VLAN hears it directly

Server replies with DHCP offer, Ack, no router needed.

Different N/w's: If it is different N/w IP's Router in between, when PC1 sends DHCP broadcast flood Router drops it, so we need to configure a helper on the Router / we need to enable Relay Agent.

49 What is the command to enable DHCP Relay agent?

configure Terminal

Interface - ethernet 0/0 -

IP helper address -> Use DHCP Server IP

Exit.